

MODEL AND DIE MATERIALS Practical

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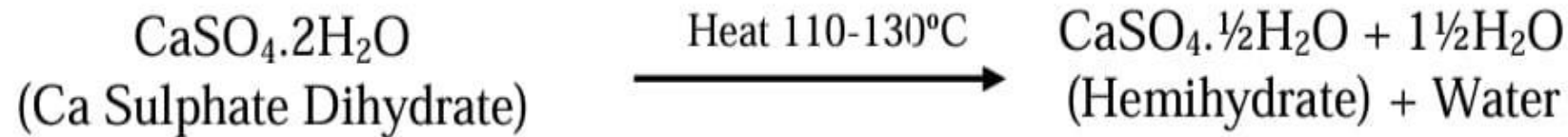
Gypsum products

It is found naturally in the form of Calcium Sulphate dihydrate ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$)

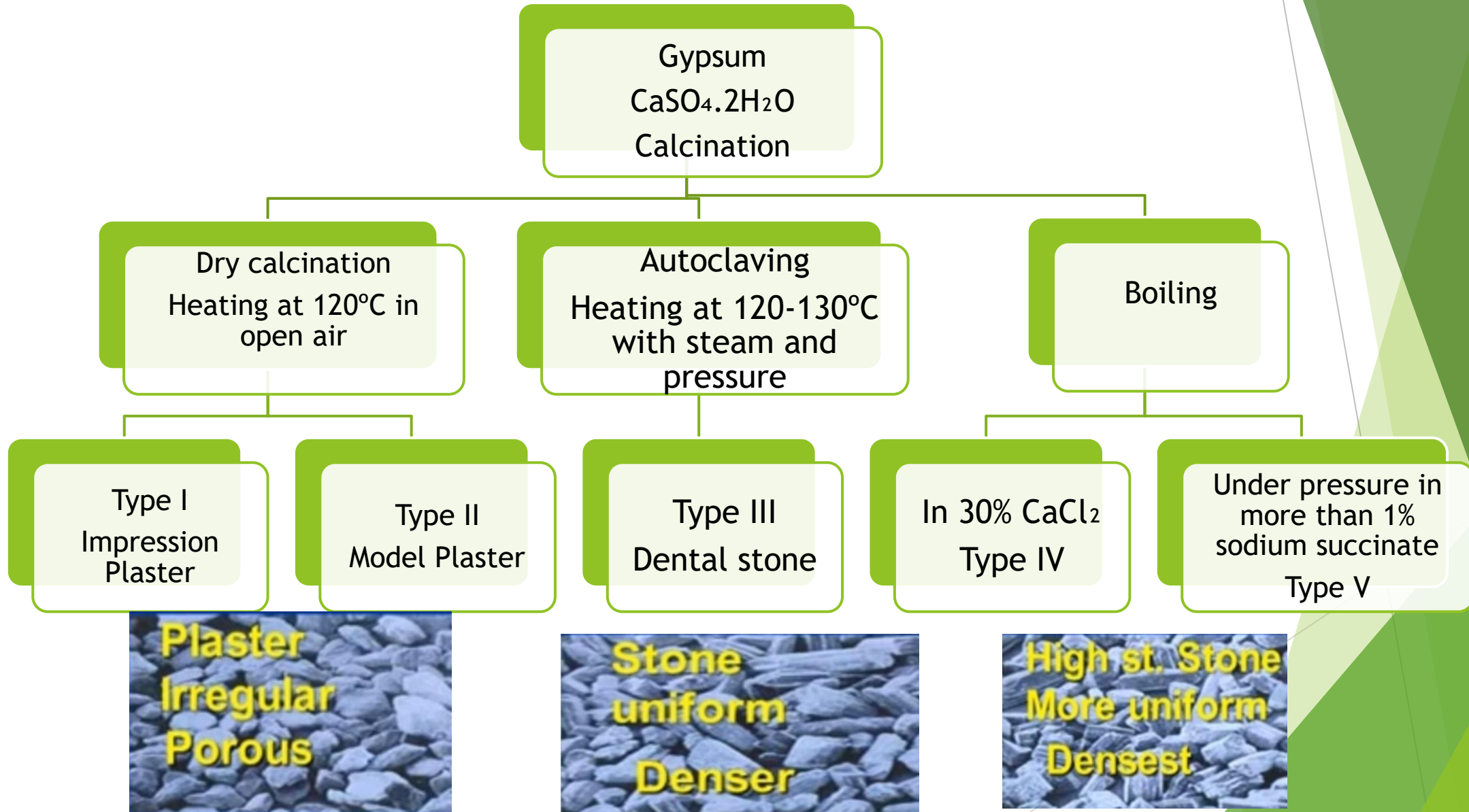
- All the forms have the same chemical composition .They differ in the method of manufacturing So they differ in properties and uses.

On heating (Calcination reaction)

Calcium Sulphate dihydrate is converted to Calcium Sulphate hemihydrate.



Manufacturing of different types



Gypsum products

- ▶ The amount of water of crystallization (essential water)(required for the chemical reaction) for different products is the same (each 100gm of $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$ requires 18.6 g of water).
- ▶ Practically ,excess water is needed why?? to form homogenous workable mix . This excess water differs among the different gypsum products.

Gypsum products

The amount of **water required** (workable mass) for 100g :

- ▶ Model plaster=55g
- ▶ Dental stone=30g
- ▶ High strength stone=22g

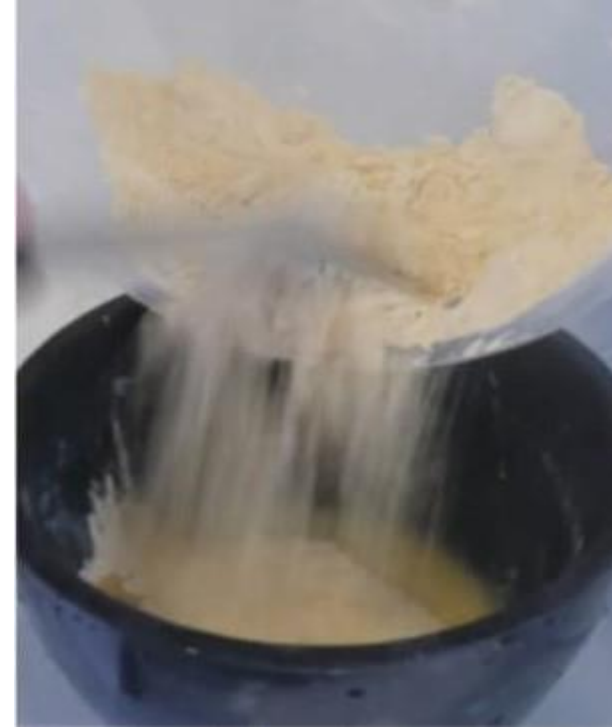
Manipulation

- ▶ The suitable water/powder ratio is proportioned.
- ▶ Mixing is done using rubber bowl and rigid wide spatula.



Manipulation

- ▶ Water is dispensed first then the powder is added.
- ▶ The powder is allowed to settle for about 30 seconds to reduce air incorporation.



Manipulation

- Mix vigorously with wiping the inside surface of the bowl until obtaining homogenous mix (about 1 minute).



Manipulation

- ▶ Vibrate the mixing using a vibrator
or by tapping the bowl against the bench.

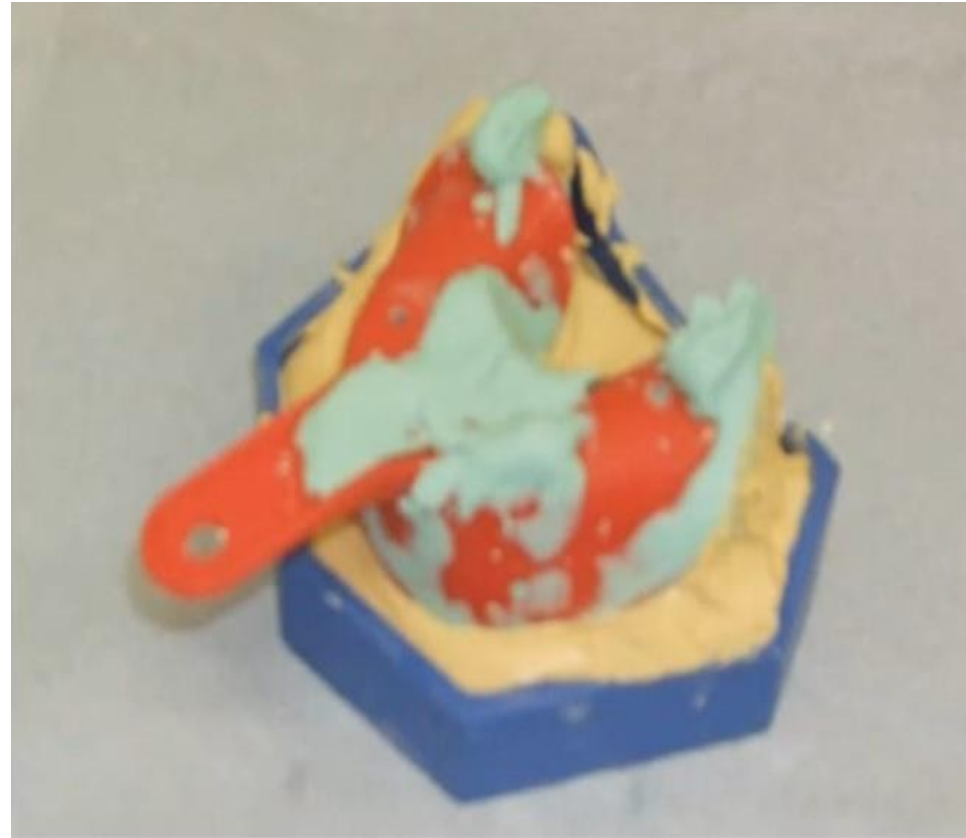


Manipulation

- Pour the impression with care to entrap no air bubbles.



Manipulation



Manipulation

- After 1 hour, separate the impression from the cast.



Manipulation

- ▶ Mechanical mixers are available.
- ▶ They mix the Gypsum under vacuum.



Manipulation

- ▶ Be sure that the bowl and spatula **should be** clean. Remnants from previous mixing accelerate the setting reaction by acting as nuclei of crystallization.
- ▶ **N.B.** Gypsum products powder should be protected from moisture contamination.
- ▶ During storage because moisture contamination may change some hemihydrate particles into dihydrate. This will **increase the number of nuclei of crystallization** leading to **decrease in the setting time.**
- ▶ But, after a while more and more dihydrate is formed leading to reverse effect and increase in setting time.

Properties

1- setting time:

- It is the time needed for completion of the setting reaction, which is divided into:

Time	from	until	minutes
Mixing time	Add powder to water	Obtain homogenous workable mix.	1 minute
Working time		Having even consistency (workable mix).	3 minutes
Initial setting		Material not flow . It can be carved.	12 minutes
Ready to use		Can separate the impression.	60 minutes
Final setting		Complete the reaction.	7 days

Properties

1. Setting time:

Measuring setting time :



a) Loss of surface gloss:

- ▶ The mix starts to lose its gloss. It indicates initial setting. The impression shouldn't be removed at this stage. (Estimate the proper working time).

b) Temperature rise :

- ▶ There is an increase in the temperature as the reaction is exothermic (not accurate method).

Properties

1.Setting Time:

C) Penetration test :

- ▶ It is the accurate method to measure the setting time.
- ▶ As the material set, it will resist the needle.

Vicate test (measure the initial setting time).

Gillmore test (measure the initial and final setting).

Properties

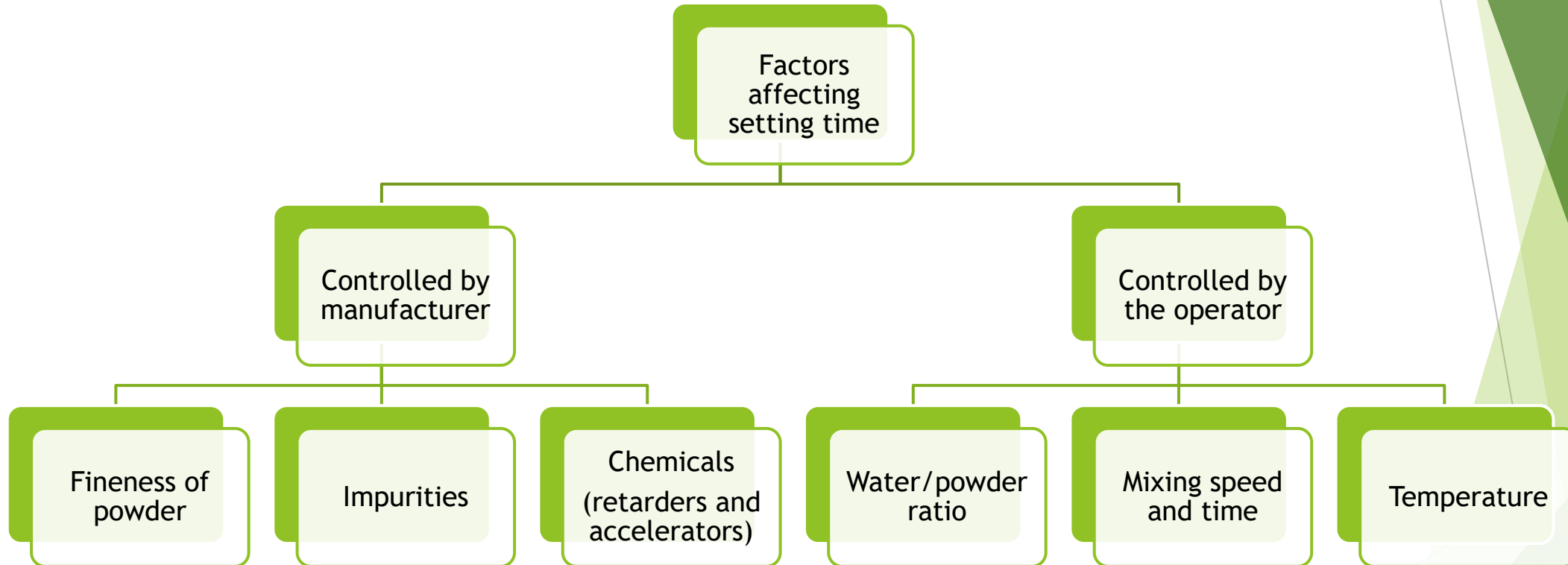


Vicate



Gillmore

Properties



Properties

1. Setting Time:

Factors controlled by the operator

a) Water / Powder ratio:

- ▶ High W/P = watery mix → the less the number of nuclei of crystallization → retardation
- ▶ Low W/P = thick mix → acceleration

Properties

- ▶ Such excess water will be present in the final product as **free water**.
- ▶ This excess water will evaporate ,leaving the set material porous.
- ▶ It may take 7 days to lose the excess water.
- ▶ Thus, the **more water** used in mixing, the more porous the set product, the less hard and the weaker the gypsum will be.

Properties

1. Setting Time:

Factors controlled by the operator

b) Mixing speed and time(within limit):

- ▶ Increasing in spatulation (speed or time or both)  Accelerate the reaction (breaking up some dihydrate crystals so increasing the number of nuclei of crystallization).

Properties

1. Setting Time:

Factors controlled by the operator

c) Water temperature:

- ▶ 20:50 °C → accelerate the reaction
- ▶ Above 50 °C → retard the reaction
- ▶ 100 °C → no reaction (no setting occurs as solubility of hemihydrate = solubility of Dihydrate)

Care of cast

- ▶ The safest method for soaking the cast is to place the cast in a water bath with gypsum debris remaining material located at the bottom of the container (saturated solution of calcium sulphate).

Disinfection or Infection control

- ▶ Models can be disinfected by immersion in 1:10 dilution of sodium hypochlorite to prevent cross contamination to the dental office personnel by micro-organisms including Hepatitis B Virus and Human Immuno deficiency Virus(HIV).

Uses

Depending on the properties of each product, they have different uses.

1. Impression plaster "Type I":

Rarely used nowadays as it is anelastic, has low mechanical properties and its setting reaction is exothermic. Used for mounting articulators, and for final impression.

2. Model plaster "Type II":

Used when mechanical properties are not of prime importance as in primary casts. It is used principally for primary casts or study casts.

3. Dental stone "Type III":

Used for construction of secondary casts e.g. casts used in the fabrication of full dentures.

Uses

6.4. High strength stone "Type IV" or die stone:

It is stronger, harder and denser, therefore, used for the construction of **harder and stronger models and single dies** (figure 11c). Characterized by slight expansion.

6.5. High strength/high expansion stone "Type V":

The improved strength is attained by lowering the W/P ratio even more than Type IV stone.

In addition, the setting expansion has been increased.

It is used when **higher die expansion is needed to compensate for the solidification shrinkage of new base metal alloys having greater casting shrinkage.**

Can not be used for fabrication of inlay or onlay.



Mounting articulators: type 1 plaster

Uses

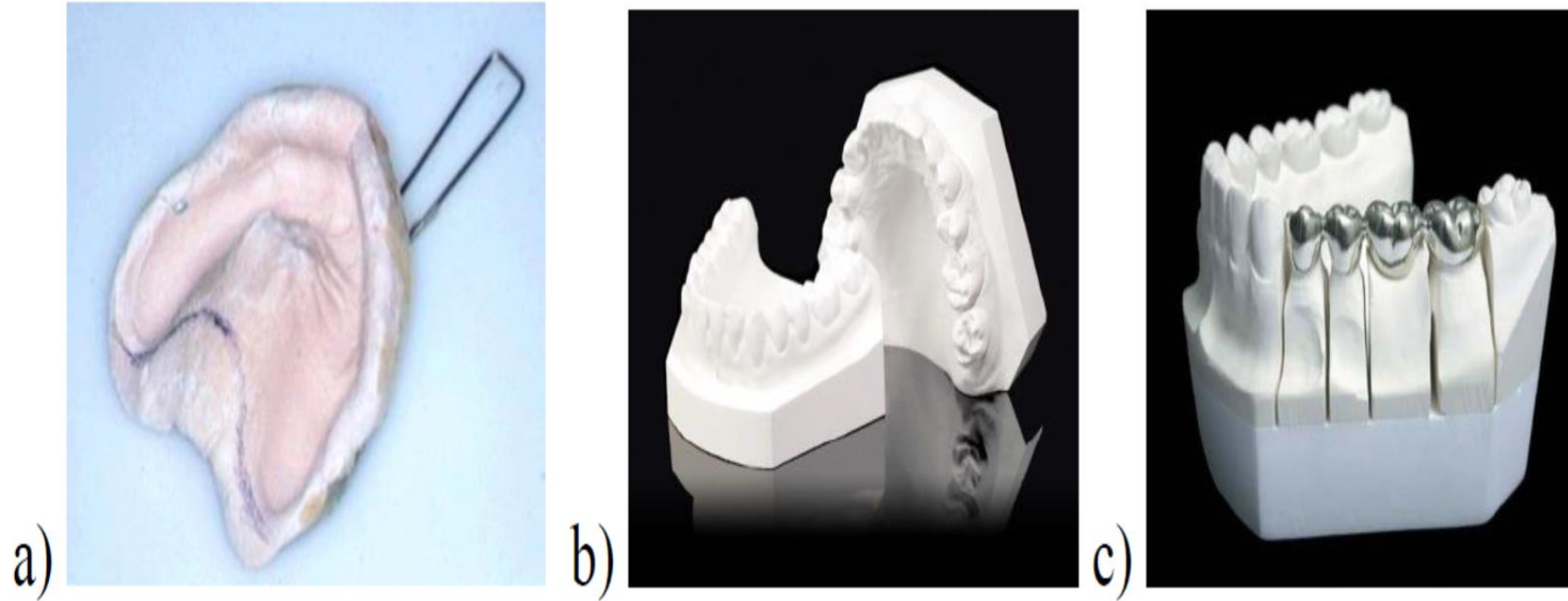


Figure 11. Uses of different types of gypsum a) Type I b) Type II c) Type IV





Thank You